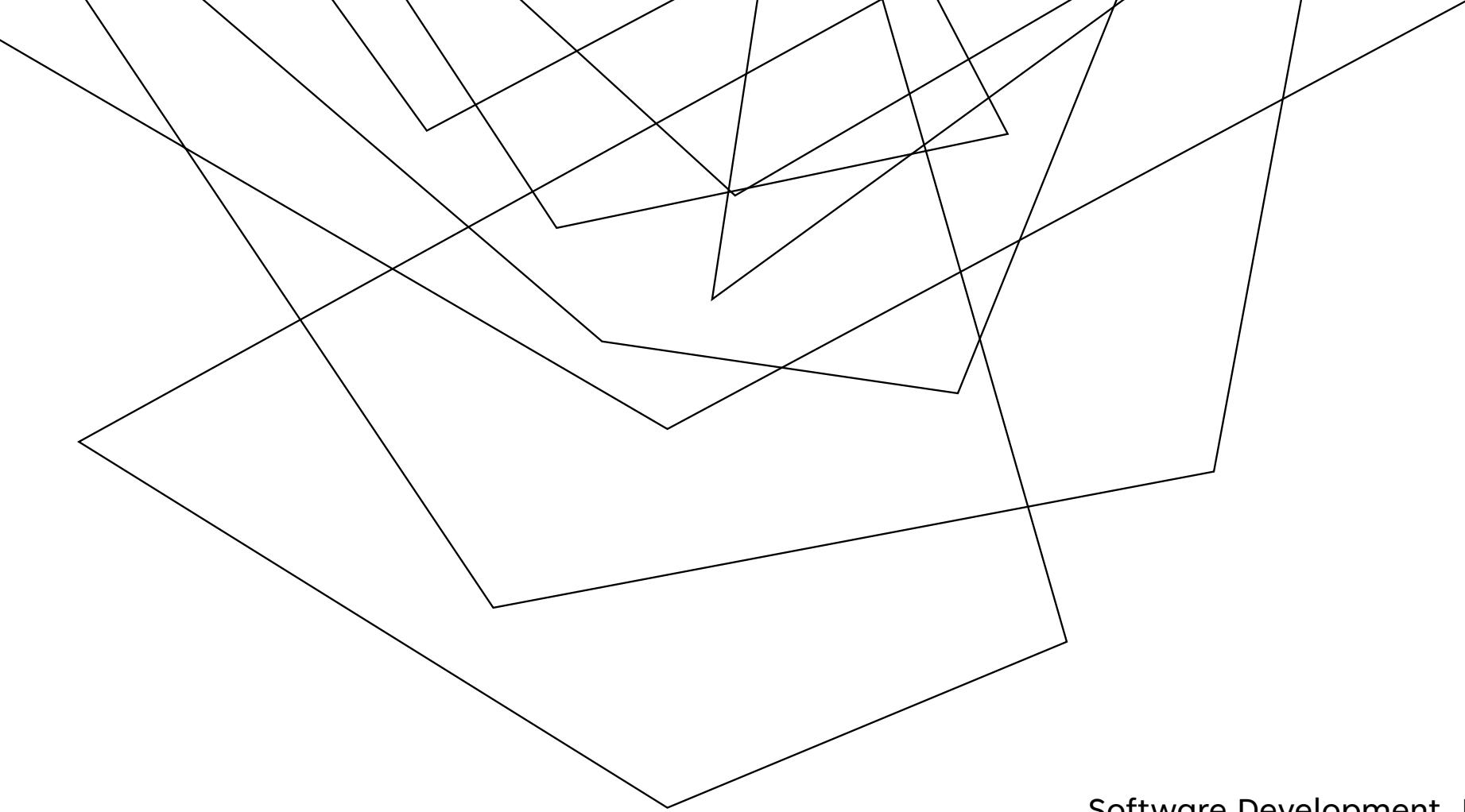


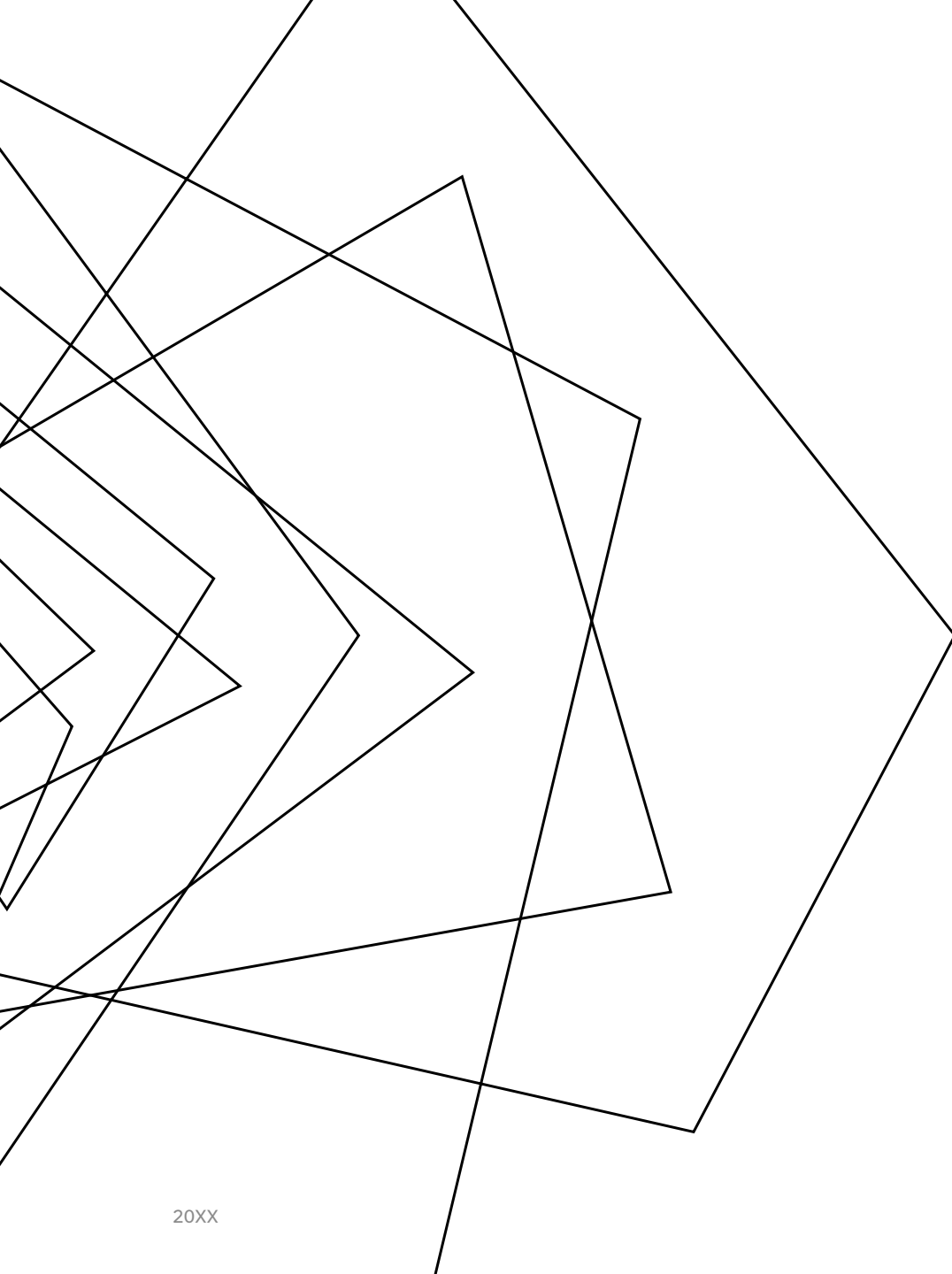


Software Development, Maintenance & Operations

# GROUP WORK PROJECT

Mikko Mäkitalo

[mikko.makitalo@oulu.fi](mailto:mikko.makitalo@oulu.fi)



# INTRODUCTION

## MIKKO MÄKITALO

Doctoral researcher focusing on teaching

PhD topic: Quantum computing (currently: Quantum-encoded Fourier signals, Bell states, and information security)

Master of Science (Computer science)

Master of Arts (Musicology)

Also some studies in:

Quantum physics, acoustics, signal processing,  
mathematics, art history, philosophy



## PROJECT INTRODUCTION: LEGACY SYSTEM MODERNIZATION FOR PQC & LLM-ASSISTED DEVELOPMENT

You will modernize (develop, maintain, and operate) a small edge–cloud system (simulation model)

Transition to post-quantum cryptography (ML-KEM)

Critically evaluate LLM-generated artifacts

Work in groups of 5 (it says 4-5 in the instructions, but it is 5)



# GENERATING SETUP & DEVELOPMENT

AI-assisted development:

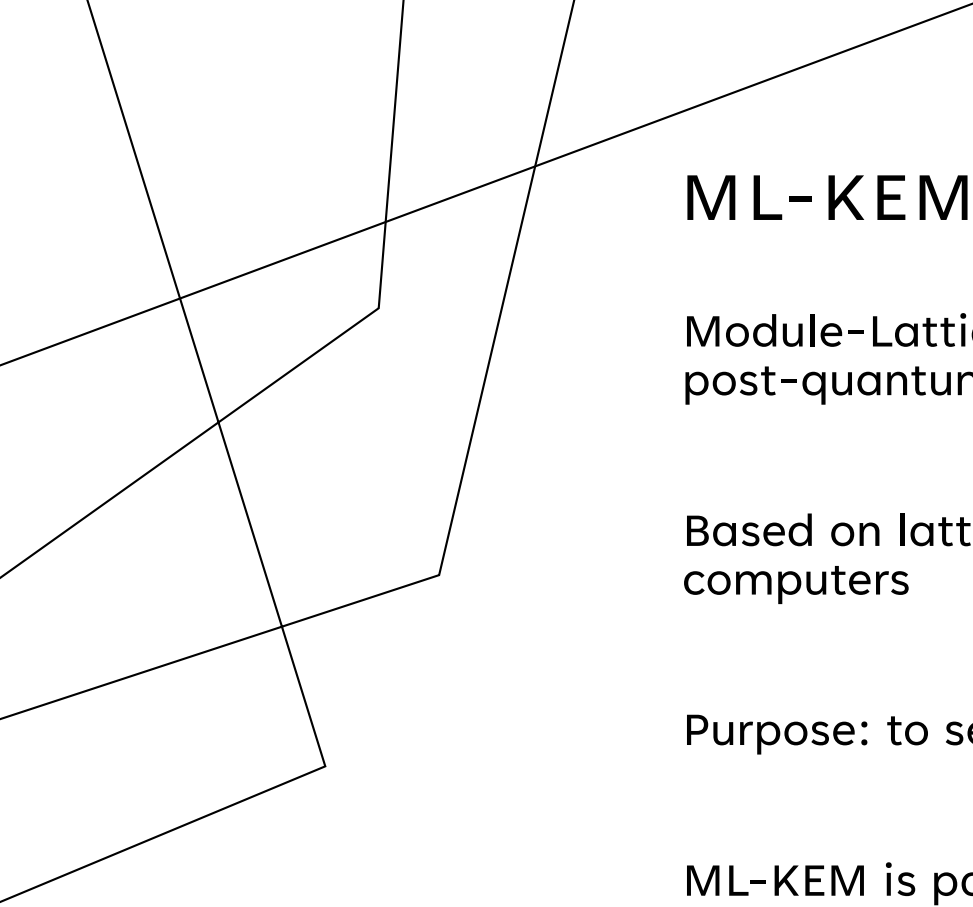
Generate selected components with an LLM and record the relevant prompts, outputs, and decisions

Baseline analysis:

Understand and document the initial architecture, run the system, and establish baseline tests and measurements

Critical evaluation and maintenance:

Test, review, correct, and improve the generated artefacts; document how problems were discovered and verified



# ML-KEM

Module-Lattice-based Key Encapsulation Mechanism, a post-quantum-secure key exchange algorithm

Based on lattice cryptography, which remains hard even for quantum computers

Purpose: to securely establish a shared secret key between two parties

ML-KEM is part of NIST's PQC standardization, known in practice as Kyber

Replaces classical methods like RSA, which are vulnerable to quantum attacks

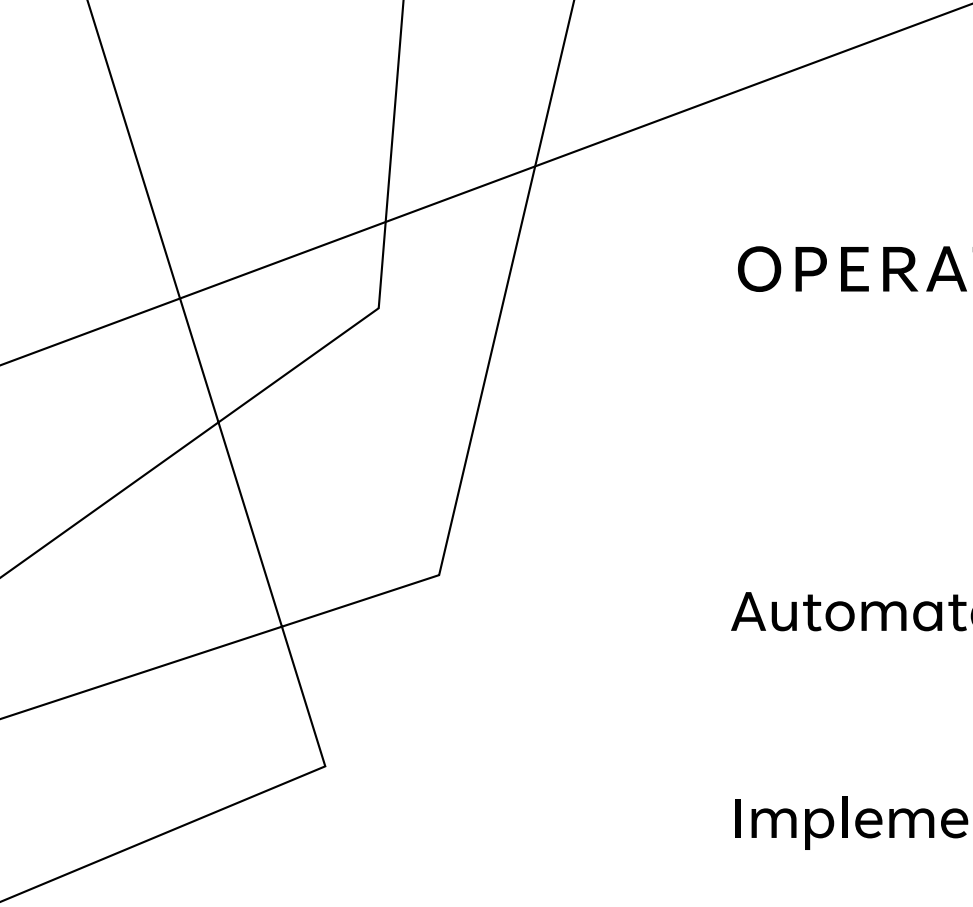
It's important to understand the basics of ML-KEM and quantum computing on a high conceptual level. There's **NO** need to understand quantum mechanics or, for example, Shor's algorithm deeply in this course



## MAINTENANCE & ML-KEM

ML-KEM integration: Integrate ML-KEM using an existing implementation (Python libraries)

Preserve legacy (for example Arduino) compatibility where appropriate and provide a clearly justified migration and fallback strategy



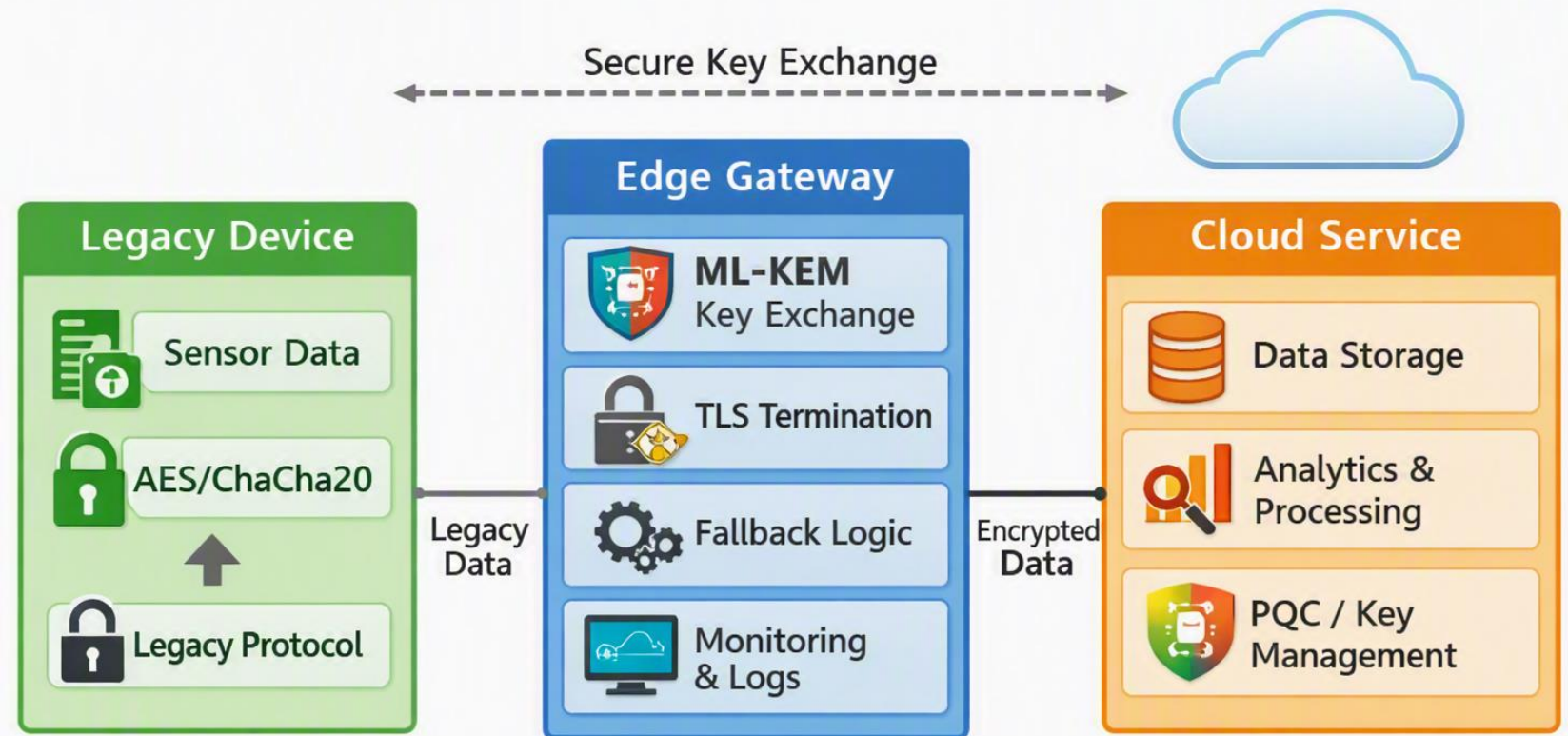
# OPERATIONS

Automate build and testing

Implement deployment to a test environment

Add health checks, logs, and metrics

# Edge-Cloud PQC Architecture





# FINAL EVALUATION & FINALIZING THE PROJECT

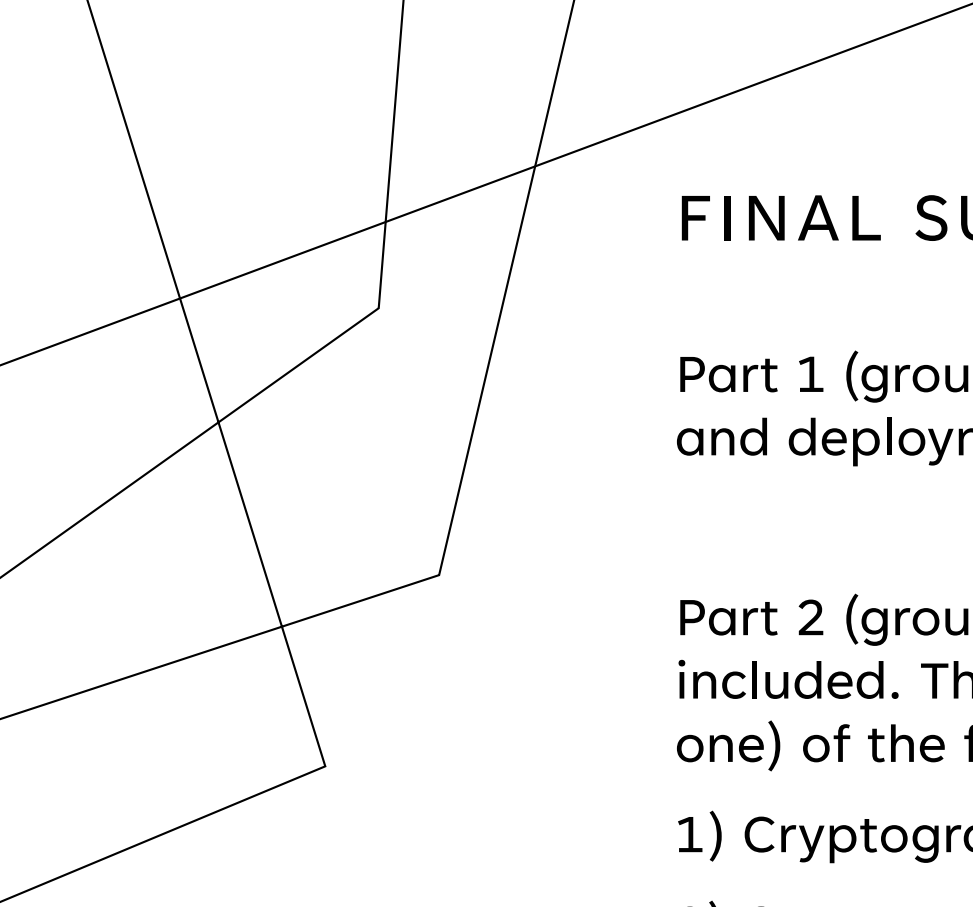
Analyze and demonstrate strengths and weaknesses

Identify remaining technical debt and risks

Peer review

Presentation

Fine-tuning and wrapping it up



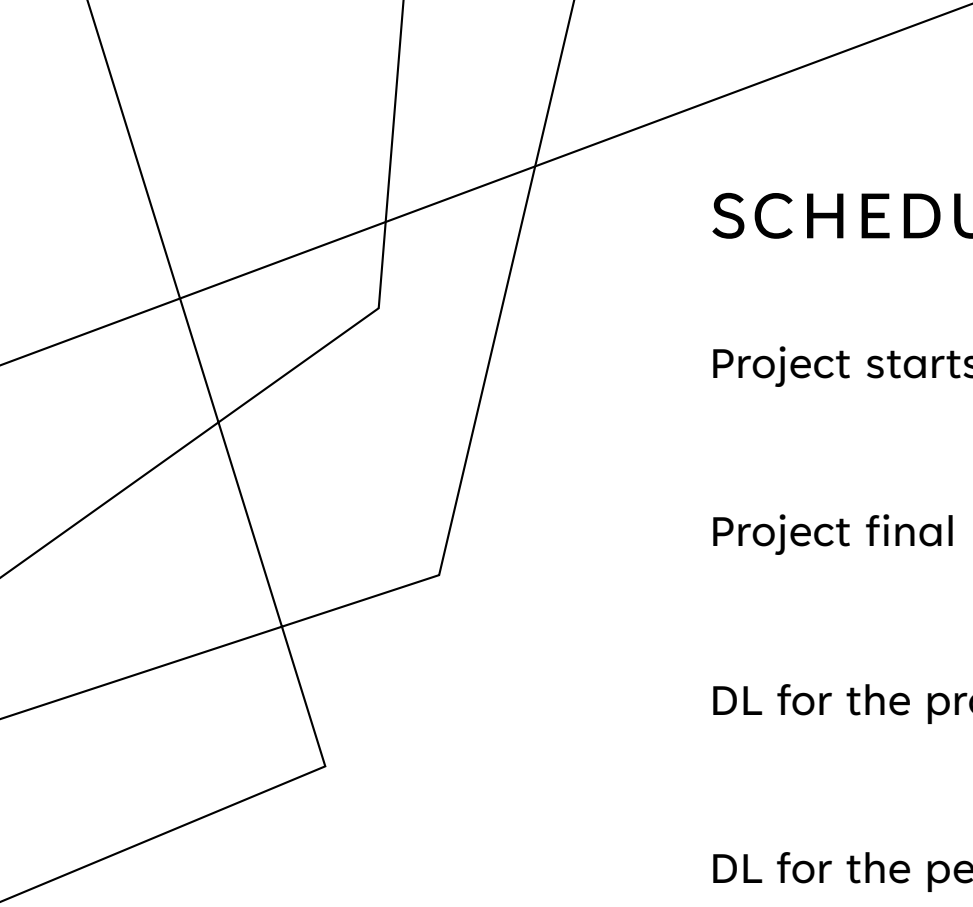
# FINAL SUBMISSION

Part 1 (group): Technical documentation, source code (Python), CI/CD and deployment configuration

Part 2 (group): Project report (max. 5 pages). References must be included. The report should examine the project from one (choose only one) of the following three perspectives:

- 1) Cryptography and mathematics
- 2) Quantum threats and attackers
- 3) Information security and organizational cybersecurity

Part 3 (individual): Reflection (400-500 words)



# SCHEDULE & IMPORTANT DATES

Project starts: Mon 7th Sep

Project final DL: Sun, 25th Oct

DL for the project submission for the peer reviews: Sun, 11th Oct

DL for the peer reviews to be ready: Wed, 14th Oct

Presentations: 15th and 16th Oct (Thursday and Friday)

Q&A sessions: hybrid or online every Tuesday, 15.9. - 13.10.2026



## GOOD LUCK & TAKE IT EASY

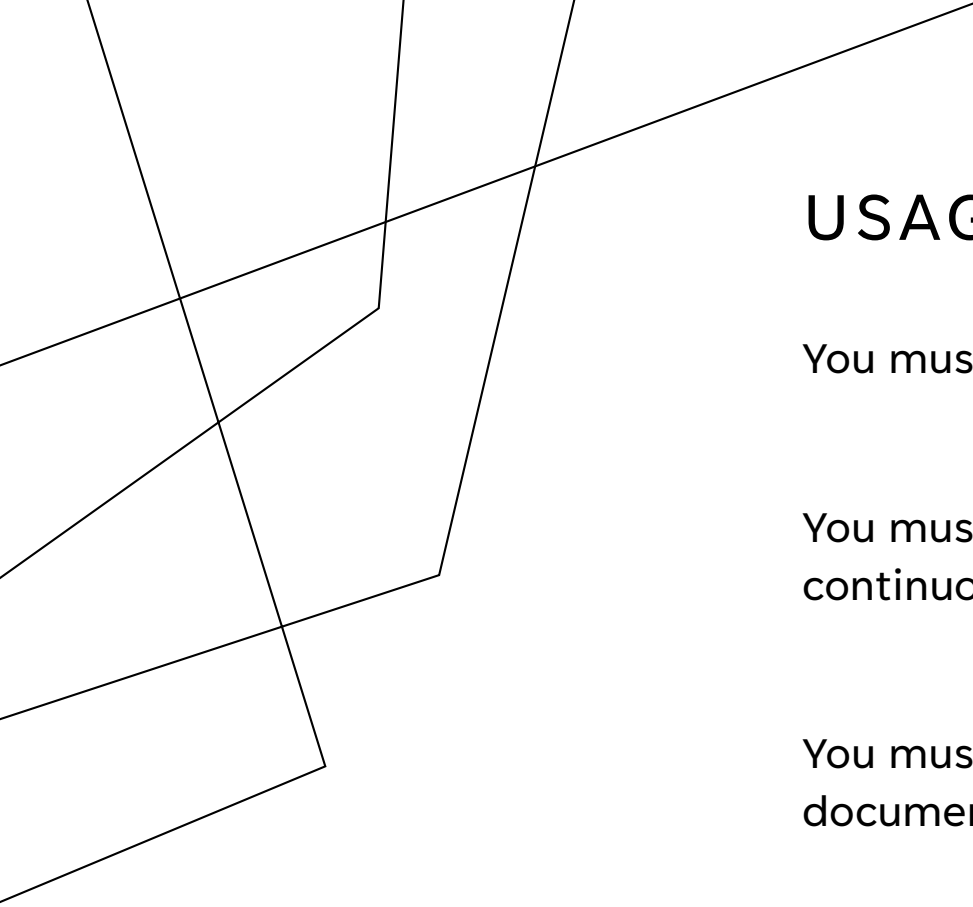
Be creative! The instructions are loose on purpose. If you think something is not specified in detail, then it probably isn't 😊

Attend the Q&A sessions

Start working immediately and form the groups

Read the project description on Moodle once again carefully so that you understand what the project is all about

Keep it simple! Please do **NOT** try to make the project any bigger than needed. It's a good thing to keep it plain and simple



# USAGE OF AI

You must use AI to create the code (part 1)

You must manually evaluate the AI-generated code continuously (part 1)

You must include the prompts in the technical documentation file (part 1)

You must not use AI when you write your personal reflections, experiences, and feelings (part 3)

You must report how and where AI was used



THANK YOU!

Mikko Mäkitalo

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